

PHASE – 1 DELIVERABLES

Smart Electric Vehicle Battery Health Monitoring System

Prototype Platform: Arduino Uno

1. Problem Statement

Electric vehicle (EV) batteries are critical for vehicle safety, driving range, and performance. Problems such as over-voltage, under-voltage, excessive current, overheating, and battery degradation can reduce battery life and may create unsafe operating conditions. Conventional vehicle indicators may not provide sufficient information about battery condition for a simple educational monitoring prototype. Therefore, a low-cost embedded system is proposed to continuously monitor important battery parameters and provide early warnings when abnormal conditions occur. The proposed system uses an Arduino Uno to collect sensor data, process the readings, display battery status, and activate an alert when a preset limit is exceeded.

2. Abstract (200–300 words)

The Smart Electric Vehicle Battery Health Monitoring System is an embedded monitoring prototype designed to improve awareness of electric vehicle battery condition and safety. The system uses an Arduino Uno as the main controller and continuously monitors important parameters such as battery voltage, charging/discharging current, and battery temperature. These parameters are obtained using suitable voltage, current, and temperature sensing circuits. The Arduino Uno processes the sensor readings and compares them with predefined safe operating limits. The system provides real-time information through an LCD/OLED display and can use LEDs or a buzzer to indicate abnormal conditions. If the measured voltage becomes too high or too low, current exceeds a selected limit, or temperature rises above a safe threshold, the system generates an immediate warning. A simple State of Charge (SoC) estimation can also be included using battery voltage as an educational prototype method. For a more advanced implementation, battery State of Health (SoH) can be estimated using historical capacity, cycle count, internal resistance, and data-analysis methods. The proposed prototype is intended for academic demonstration and laboratory testing rather than direct connection to a high-voltage EV battery pack. Appropriate isolation, protection, fusing, and a low-voltage battery source should be used during experiments. The system can be extended in the future with wireless communication, IoT/cloud storage, mobile notifications, and improved battery-health algorithms. Overall, the project provides a practical and low-cost approach to understanding EV battery monitoring and early fault detection.

3. Objectives

- Develop a low-cost EV battery monitoring prototype using Arduino Uno.
- Measure battery voltage, current, and temperature in real time.
- Display the measured parameters clearly to the user.
- Provide early warning for abnormal voltage, current, and temperature.
- Implement a basic prototype-level State of Charge (SoC) indication.
- Study battery health parameters and identify possible degradation indicators.
- Provide a foundation for future IoT and data-logging expansion.

4. Scope of the Project

- Arduino Uno based sensing and monitoring.
- Low-voltage prototype battery testing.
- Voltage, current and temperature measurement.
- LCD/OLED or serial-monitor display.
- Buzzer/LED based abnormal-condition alert.
- Basic SoC estimation for demonstration.
- Future expansion to IoT, cloud monitoring, data logging and advanced SoH estimation.

5. Expected Outcomes

- A working Arduino Uno based battery monitoring prototype.
- Real-time display of voltage, current and temperature.
- Warning when measured values cross configured thresholds.
- Basic battery charge-status indication.
- Improved understanding of EV battery safety and monitoring.
- A scalable platform for future IoT and intelligent battery-health analysis.

6. Literature Survey

Area	Common Approach	Limitation / Gap
Battery Voltage Monitoring	Voltage sensor/divider and microcontroller	Voltage alone does not fully describe battery health
Current Monitoring	Hall-effect/shunt current sensing	Requires calibration and noise filtering
Temperature Monitoring	Thermistor/DS18B20/temperature IC	Sensor placement affects accuracy
SoC Estimation	Voltage-based or coulomb counting methods	Simple voltage methods can be inaccurate under load
SoH Estimation	Capacity, resistance and cycle-data analysis	Needs historical data and better algorithms
IoT Battery Monitoring	Wireless communication and cloud dashboard	Adds communication and power requirements

7. Existing Solutions Analysis

Commercial EV Battery Management Systems (BMS) provide sophisticated functions such as cell-level voltage monitoring, temperature sensing, balancing, protection, State of Charge (SoC) estimation, State of Health (SoH) estimation, and fault management. However, they can be complex and costly for an academic prototype. A simple Arduino Uno system can demonstrate the basic monitoring and alert functions using low-voltage test batteries.

8. Gap Analysis

Parameter	Typical Basic Monitoring	Proposed Prototype
Voltage	Measured/displayed	Measured + threshold alert
Current	Sometimes measured	Measured + abnormal-current alert
Temperature	May be monitored	Continuous monitoring + warning
SoC	Basic indicator	Prototype voltage-based estimation
SoH	Often unavailable	Study/plan for future data-based estimation
User Alert	Dashboard dependent	LCD + LED/buzzer
Connectivity	Depends on product	Optional future IoT extension

9. Proposed System Architecture

Low-Voltage Battery → Voltage Sensor + Current Sensor + Temperature Sensor → Arduino Uno → Processing / SoC Estimation → LCD/OLED Display + LED/Buzzer Alert

Recommended prototype components: Arduino Uno, suitable voltage divider/sensor module, Hall-effect current sensor such as ACS712 for an appropriate low-current test setup, DS18B20 or another suitable temperature sensor, 16×2 I2C LCD or OLED, buzzer, LEDs and resistors. Do not connect an Arduino directly to a high-voltage EV battery pack.

10. Arduino Uno Program Concept

The program initializes the sensors and display, reads voltage/current/temperature values repeatedly, compares them with configured thresholds, displays the readings, and activates the warning output when an unsafe condition is detected. A prototype SoC percentage can be mapped from a calibrated battery-voltage range. The exact thresholds must be selected according to the low-voltage battery chemistry and test setup.

11. References

- Arduino Uno documentation and technical specifications.
- Battery Management System (BMS) textbooks and technical literature on Li-ion battery monitoring.
- Datasheets for the selected voltage sensor, current sensor and temperature sensor.
- Relevant IEEE/academic papers on EV battery State of Charge (SoC) and State of Health (SoH) estimation.
- Manufacturer safety documentation for the selected prototype battery.

12. Git Installation & GitHub Learning Plan

- Install Git and configure username/email.
- Create a GitHub repository for the project.
- Add the Arduino source code, circuit diagram, documentation and test results.
- Use commits to record each development stage.
- Use branches for major changes and merge tested versions.
- Maintain a README containing project objective, components, wiring, program steps and results.

Safety Note: This document describes an educational prototype. Use only an appropriate low-voltage battery for laboratory testing. High-voltage EV battery packs can cause severe injury or fire and require professional BMS, isolation, protection and test procedures.